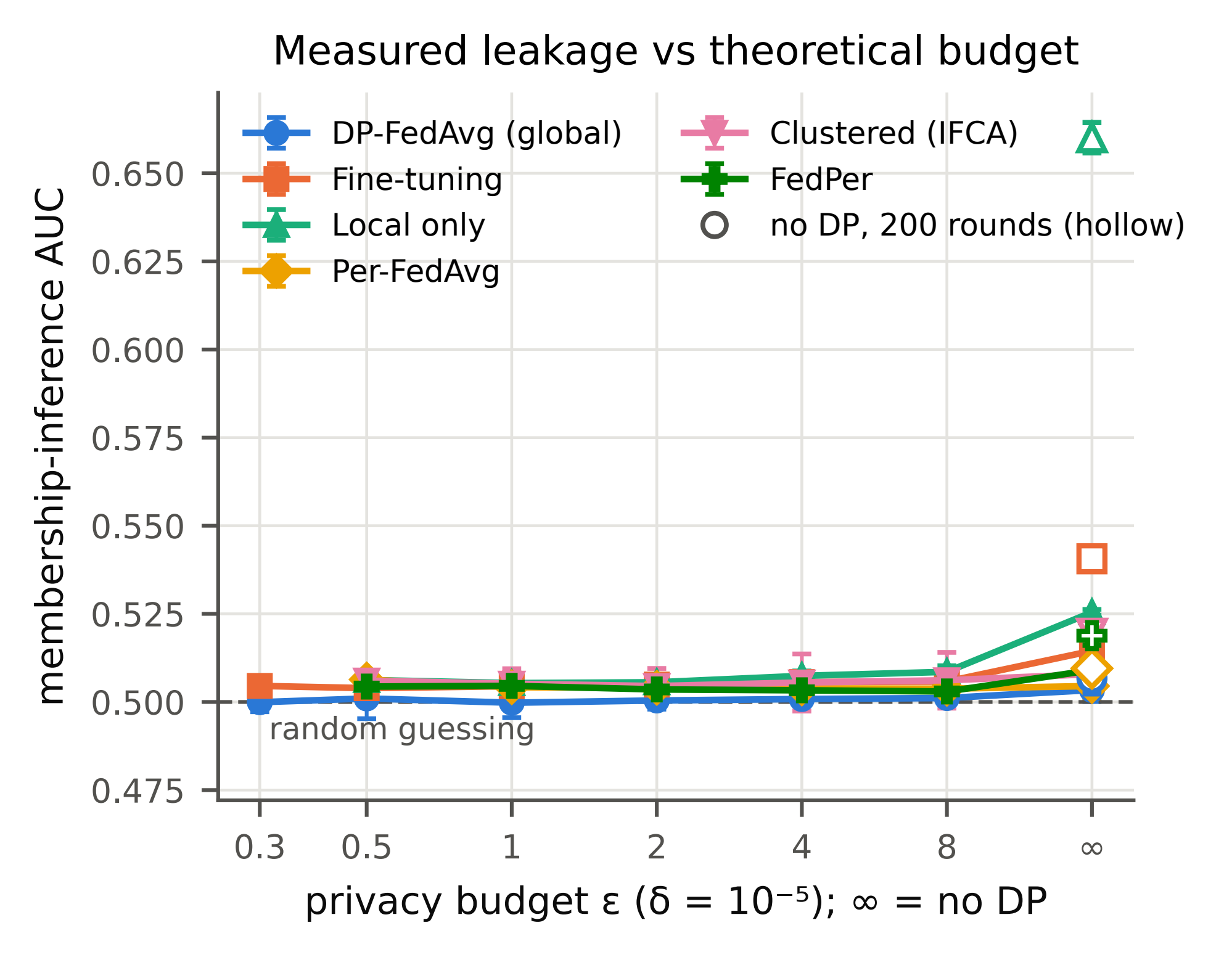
Measured leakage vs theoretical budget
DP-FedAvg (global)
Fine-tuning
Local only
Per-FedAvg
Clustered (IFCA)
FedPer
no DP, 200 rounds (hollow)
membership-inference AUC
0.650
0.625
0.600
0.575
0.550
0.525
0.500
0.475
random guessing
0.3
0.5
1
2
4
8
∞
privacy budget ε (δ = 10⁻⁵); ∞ = no DP